

School Homework Helper

Vercel Deployment Guide

Complete step-by-step guide to go live

Domain: www.schoolhomeworkhelper.com

Hosting: Vercel | DNS: Bluehost cPanel

Stack: Next.js 15 + Supabase + OpenAI

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This document walks you through every step from your local project to a live production website with HTTPS, Google Sign-In, AI tools, and your custom domain.

What You Will Accomplish

- Push your code safely to GitHub (no secrets exposed)
- Deploy the Next.js app on Vercel with one click
- Connect Supabase for login and database
- Connect OpenAI for all 6 AI tools
- Point `www.schoolhomeworkhelper.com` from Bluehost to Vercel
- Enable free HTTPS (SSL) automatically
- Verify everything works on desktop and mobile

Architecture Overview

Your domain stays registered at Bluehost. DNS records in Bluehost tell browsers to load the website from Vercel servers. Supabase and OpenAI stay as external services - you do not host a database on Vercel.

```
User browser
  |
  v
Bluehost DNS (schoolhomeworkhelper.com)
  |
  v
Vercel (Next.js app + API routes)
  |--- Supabase (auth + database)
  |--- OpenAI (AI responses)
  |--- Google OAuth (sign-in)
```

Prerequisites Checklist

- Project folder on your computer (School_homework_helper)
- GitHub account (free): <https://github.com>
- Vercel account (free): <https://vercel.com>
- Bluehost cPanel login (domain already assigned)
- Supabase project with `schema.sql` ready
- OpenAI API key with billing enabled
- Google Cloud OAuth credentials (for Sign-In)
- Node.js installed locally (for testing build)

PART 1 - Prepare Your Project Locally

Step 1: Test the build on your computer

Open terminal in your project folder and run:

```
pnpm install
pnpm run build
```

If build succeeds, your project is ready for Vercel. Fix any errors before continuing.

Step 2: Confirm .env.local is NOT uploaded to GitHub

Your .gitignore file already contains:

```
.env*.local
.env
```

Never commit API keys. You will copy values manually into Vercel dashboard.

Step 3: Note your environment variable values

Open .env.local and copy these values to a secure notepad (password manager):

- OPENAI_API_KEY
- OPENAI_MODEL (optional, default gpt-4o-mini)
- NEXT_PUBLIC_SUPABASE_URL
- NEXT_PUBLIC_SUPABASE_ANON_KEY
- SUPABASE_SERVICE_ROLE_KEY
- NEXT_PUBLIC_ADSENSE_CLIENT_ID (if using ads)
- NEXT_PUBLIC_ADSENSE_SLOT (if using ads)

For production, NEXT_PUBLIC_SITE_URL must be:

```
https://www.schoolhomeworkhelper.com
```

Step 4: Install Git (if not installed)

Download from <https://git-scm.com>. Verify with: `git --version`

PART 2 - Push Code to GitHub

Step 1: Create a new GitHub repository

- Go to github.com > click + > New repository
- Name: school-homework-helper (or your choice)
- Visibility: Private recommended (keeps code hidden)
- Do NOT add README, .gitignore, or license (project already has them)
- Click Create repository

Step 2: Initialize Git and push (first time only)

In your project folder terminal:

```
git init
git add .
git status
git commit -m "Initial commit - School Homework Helper"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/school-homework-helper.git
git push -u origin main
```

Replace YOUR_USERNAME with your GitHub username. Use GitHub login if prompted.

Step 3: Verify on GitHub

- Refresh your repo page - all project files should appear
- Confirm .env.local is NOT visible in the repo

PART 3 - Configure Supabase for Production

Step 1: Run database schema

- Login to supabase.com > your project
- Left menu: SQL Editor > New query
- Copy contents of supabase/schema.sql from your project
- Click Run
- Creates tables: users, ai_requests, blog_posts

Step 2: Enable Google provider

- Authentication > Providers > Google
- Toggle Enable
- You will add Client ID and Secret in Part 4

Step 3: Set Site URL and Redirect URLs

- Authentication > URL Configuration
- Site URL: <https://www.schoolhomeworkhelper.com>
- Redirect URLs - add ALL of these:

```
https://www.schoolhomeworkhelper.com/**  
https://schoolhomeworkhelper.com/**  
http://localhost:3000/**
```

Step 4: Copy Supabase keys

- Project Settings > API
- Project URL -> NEXT_PUBLIC_SUPABASE_URL
- anon public key -> NEXT_PUBLIC_SUPABASE_ANON_KEY
- service_role key -> SUPABASE_SERVICE_ROLE_KEY (keep secret!)

PART 4 - Configure Google Sign-In (OAuth)

Step 1: Create Google Cloud project

- Go to console.cloud.google.com
- Create project or select existing one

Step 2: Configure OAuth consent screen

- APIs & Services > OAuth consent screen
- User type: External
- App name: School Homework Helper
- Support email: your email
- Save and continue through scopes and test users if needed

Step 3: Create OAuth credentials

- APIs & Services > Credentials > Create Credentials
- Choose: OAuth client ID
- Application type: Web application
- Name: School Homework Helper Web

Step 4: Authorized JavaScript origins - add:

```
https://www.schoolhomeworkhelper.com  
https://schoolhomeworkhelper.com
```

```
http://localhost:3000
```

Step 5: Authorized redirect URIs - add Supabase callback:

Find callback URL in Supabase > Auth > Google provider. Usually:

```
https://YOUR_PROJECT_ID.supabase.co/auth/v1/callback
```

Step 6: Paste into Supabase

- Copy Google Client ID and Client Secret
- Paste into Supabase > Auth > Google provider
- Save

PART 5 - Deploy on Vercel

Step 1: Create Vercel account

- Go to vercel.com
- Sign up with GitHub (recommended - easiest import)

Step 2: Import your GitHub repository

- Dashboard > Add New > Project
- Find school-homework-helper repository
- Click Import

Step 3: Configure project settings

- Framework Preset: Next.js (auto-detected)
- Root Directory: ./ (leave default)
- Build Command: next build (default)
- Output Directory: .next (default)
- Install Command: npm install or pnpm install

Step 4: Add Environment Variables BEFORE first deploy

Expand Environment Variables section. Add each variable:

- OPENAI_API_KEY = sk-... [Production, Preview, Development]
- OPENAI_MODEL = gpt-4o-mini [Production (optional)]
- NEXT_PUBLIC_SITE_URL = <https://www.schoolhomeworkhelper.com> [Production only]
- NEXT_PUBLIC_SUPABASE_URL = <https://xxx.supabase.co> [All environments]
- NEXT_PUBLIC_SUPABASE_ANON_KEY = eyJ... [All environments]
- SUPABASE_SERVICE_ROLE_KEY = eyJ... (secret) [Production only]
- NEXT_PUBLIC_ADSENSE_CLIENT_ID = ca-pub-... [Production (optional)]
- NEXT_PUBLIC_ADSENSE_SLOT = your slot id [Production (optional)]

IMPORTANT: NEXT_PUBLIC_SITE_URL must be your live domain in Production.

Step 5: Click Deploy

- Wait 2-5 minutes for build to complete
- Green checkmark = success
- You get a temporary URL: [your-project.vercel.app](#)
- Open it and test homepage + one AI tool

Step 6: If build fails

- Click failed deployment > View Build Logs
- Common fixes: missing env vars, TypeScript errors
- Run `pnpm run build` locally to reproduce and fix

PART 6 - Connect Custom Domain (Bluehost DNS)

Step 1: Add domains in Vercel

- Project > Settings > Domains
- Add: schoolhomeworkhelper.com
- Add: www.schoolhomeworkhelper.com
- Vercel shows Invalid Configuration until DNS is updated

Step 2: Open Bluehost DNS Zone Editor

- Login to Bluehost
- Go to cPanel
- Domains section > Zone Editor (or DNS Zone Editor)
- Select domain: schoolhomeworkhelper.com

Step 3: Add DNS records

Add or UPDATE these records (remove conflicting old ones):

```
Type: A
Host: @
Points to: 76.76.21.21
TTL: 14400 (or 4 hours)

Type: CNAME
Host: www
Points to: cname.vercel-dns.com
TTL: 14400
```

If an old A record for @ points to Bluehost IP, delete or replace it.

Step 4: Wait for DNS propagation

- Usually 15 minutes to 2 hours; can take up to 48 hours
- Check status in Vercel > Domains
- When valid, status changes to Valid Configuration

Step 5: Set primary domain

- In Vercel Domains, choose primary: www.schoolhomeworkhelper.com
- Enable redirect: schoolhomeworkhelper.com -> www (recommended)
- Vercel auto-issues SSL certificate (HTTPS)

PART 7 - Post-Deploy Verification Checklist

- [] <https://www.schoolhomeworkhelper.com> loads homepage
- [] Logo appears in header and footer
- [] Navigation links work (Tools, Classes, Blog, etc.)
- [] Cookie banner shows at bottom; Accept and Reject work
- [] Privacy, Terms, Cookie Policy pages open
- [] All 6 AI tools pages load
- [] Guest can use AI tools 4 times without login
- [] Sign-in prompt appears after 4 free uses
- [] Google Sign-In works on /signin
- [] Signed-in user can use AI tools without limit

- ☐ Search icon works (Ctrl+K)
- ☐ Mobile layout looks correct on phone
- ☐ HTTPS padlock shows in browser (secure)
- ☐ Non-www redirects to www (if configured)

PART 8 - Updating Your Live Site Later

Step 1: Make changes locally

Edit code, test with: `pnpm dev`

Step 2: Commit and push to GitHub

```
git add .  
git commit -m "Describe your change"  
git push
```

Step 3: Vercel auto-deploys

Every push to main branch triggers a new deployment automatically. View progress in Vercel > Deployments tab.

Step 4: Add new environment variables

If you add new env vars in code, add them in Vercel > Settings > Environment Variables, then Redeploy.

PART 9 - Troubleshooting

Bluehost default page still shows

DNS not updated or still pointing to Bluehost server IP. Verify A record @ = 76.76.21.21 and CNAME www = cname.vercel-dns.com. Wait up to 48 hours. Clear browser cache.

Domain shows Invalid Configuration in Vercel

DNS records incorrect or not propagated yet. Use Vercel's DNS instructions exactly. Check for duplicate A/CNAME records.

Google Sign-In fails on live site

Add production URLs in Supabase redirect list. Add origins in Google OAuth console. Ensure NEXT_PUBLIC_SITE_URL is <https://www.schoolhomeworkhelper.com> in Vercel Production env.

AI tools return error or 500

Check OPENAI_API_KEY in Vercel env vars. Verify OpenAI account has credits. View Vercel > Project > Logs > Functions for error details.

Supabase auth not working

Verify all three Supabase env vars are set. Check redirect URLs in Supabase dashboard include production domain.

Ads not showing

AdSense must approve site first (can take days). User must click Accept on cookie banner. Verify NEXT_PUBLIC_ADSENSE_CLIENT_ID starts with ca-pub-.

Build fails on Vercel

Run `pnpm run build` locally. Fix TypeScript/lint errors. Ensure no missing dependencies in package.json.

Images not loading

Google avatars and YouTube thumbnails are allowed in next.config.ts. If adding new image domains, update remotePatterns and redeploy.

Quick Reference

Important URLs

- Live site: <https://www.schoolhomeworkhelper.com>
- Vercel dashboard: <https://vercel.com/dashboard>
- Supabase dashboard: <https://supabase.com/dashboard>
- Google Cloud Console: <https://console.cloud.google.com>
- Bluehost cPanel: via bluehost.com login
- GitHub: <https://github.com>

Local commands

```
pnpm dev      # Run locally
pnpm run build # Test production build
pnpm start    # Run production build locally
```

Support email

hello@schoolhomeworkhelper.com

End of Vercel Deployment Guide. Keep this PDF and your API keys secure. Regenerate this file anytime by running: `python scripts/generate-vercel-deployment-pdf.py`